

Global CO₂ Emissions

Trends, Regional Shifts, and Per Capita Inequalities

A data-driven overview of fossil-fuel CO₂ emissions from 1750 to 2024

Data: Global Carbon Budget 2025 | NASA Mauna Loa Observatory | Our World in Data

Prepared for policy analysts and climate negotiators

Atmospheric CO₂ Reaches 429 ppm — 50% Above Pre-Industrial Levels

429 ppm

Mauna Loa, Feb 2026

~280 ppm

Pre-industrial baseline

+53%

Increase over baseline

Current CO₂ is at least 1.5× greater than the natural increase observed at the end of the last ice age 20,000 years ago.

That earlier change took thousands of years; today's increase has occurred in roughly two centuries.

Ground-based and satellite measurements confirm the sharp acceleration, driven by burning coal, oil, and natural gas.

1750: ~280 ppm

1958: ~315 ppm
(Keeling begins)

2024: 429 ppm

Global Fossil Fuel CO₂ Emissions Grew Six-Fold: 6 Bt to 35+ Bt

6 Bt

1950

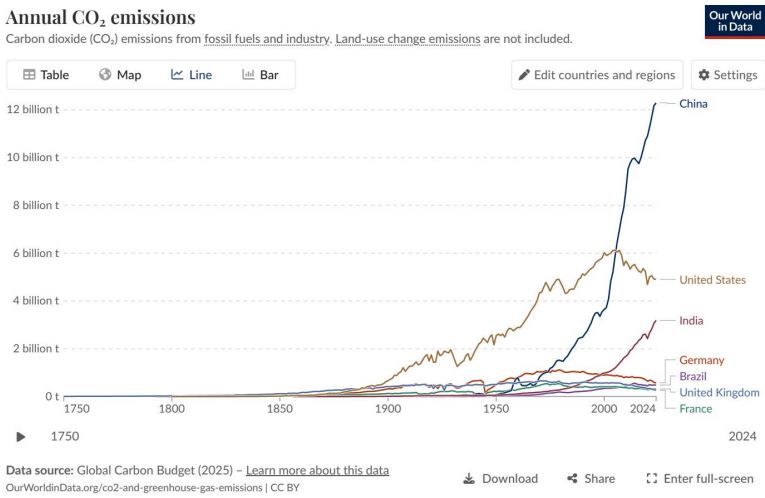
20 Bt

1990

35+ Bt

2024

Before the Industrial Revolution, CO₂ emissions were negligible. Growth remained relatively slow until the mid-20th century. Since 1950, emissions have grown more than six-fold. While growth rates have slowed in recent years, global emissions have not yet peaked.



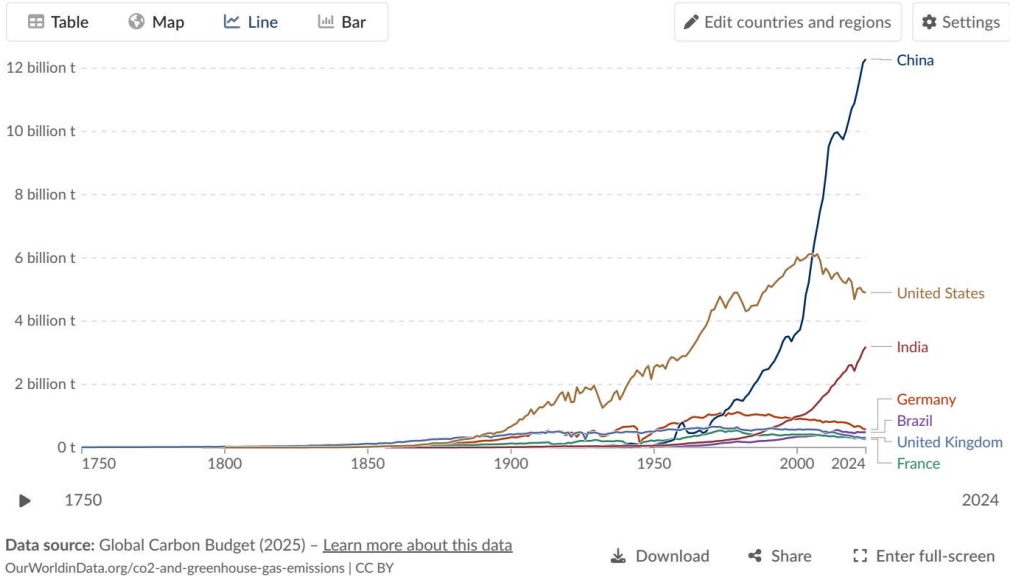
Annual CO₂ emissions by country (figure_3)

China Now Emits Over 12 Bt — More Than Double the United States

Country / Region	Annual CO ₂ (Bt)	Global Share	Trend
China	~12.5	~27%	Slight increase
United States	~5.0	~14%	Declining
India	~3.0	~8%	Rising
EU-27	~2.7	~7%	Declining
Germany	~0.6	~1.7%	Declining
United Kingdom	~0.3	~0.9%	Declining
Brazil	~0.5	~1.4%	Stable
France	~0.3	~0.8%	Declining

Annual CO₂ emissions

Carbon dioxide (CO₂) emissions from fossil fuels and industry. Land-use change emissions are not included.

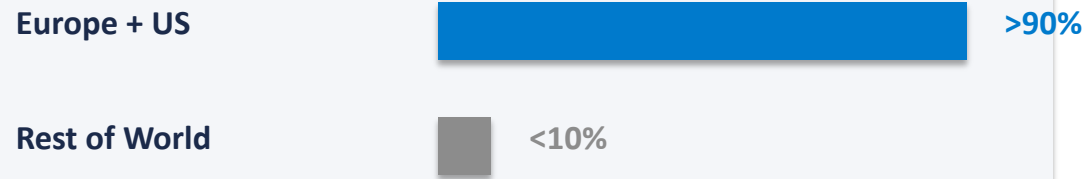


figure_3.jpg: Annual CO₂ emissions by country

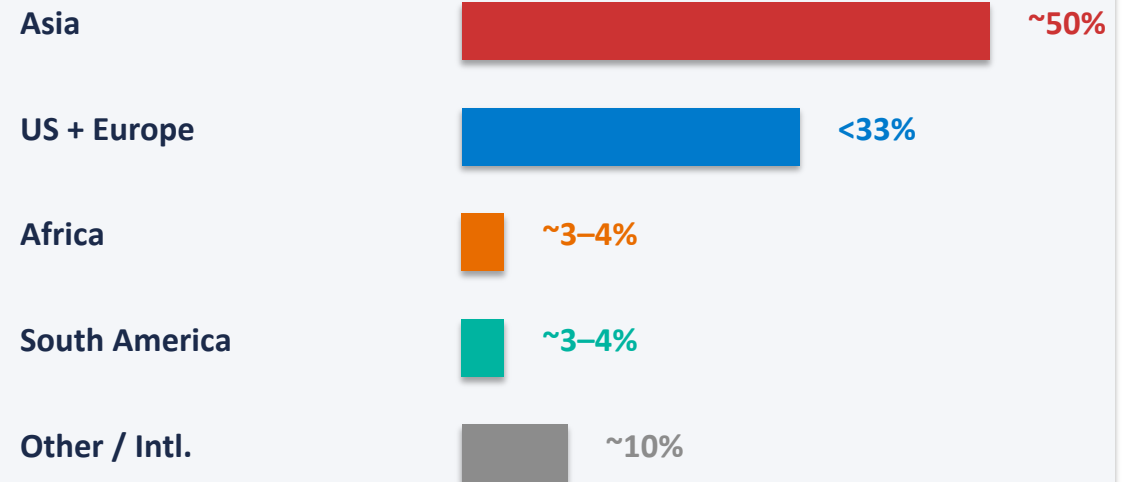
China's emissions have risen steeply since 2000. The US peaked near 6 Bt around 2005 and has since declined to ~5 Bt.

Asia Now Accounts for ~50% of Global Emissions; US + Europe Below One-Third

1900: Emission Shares

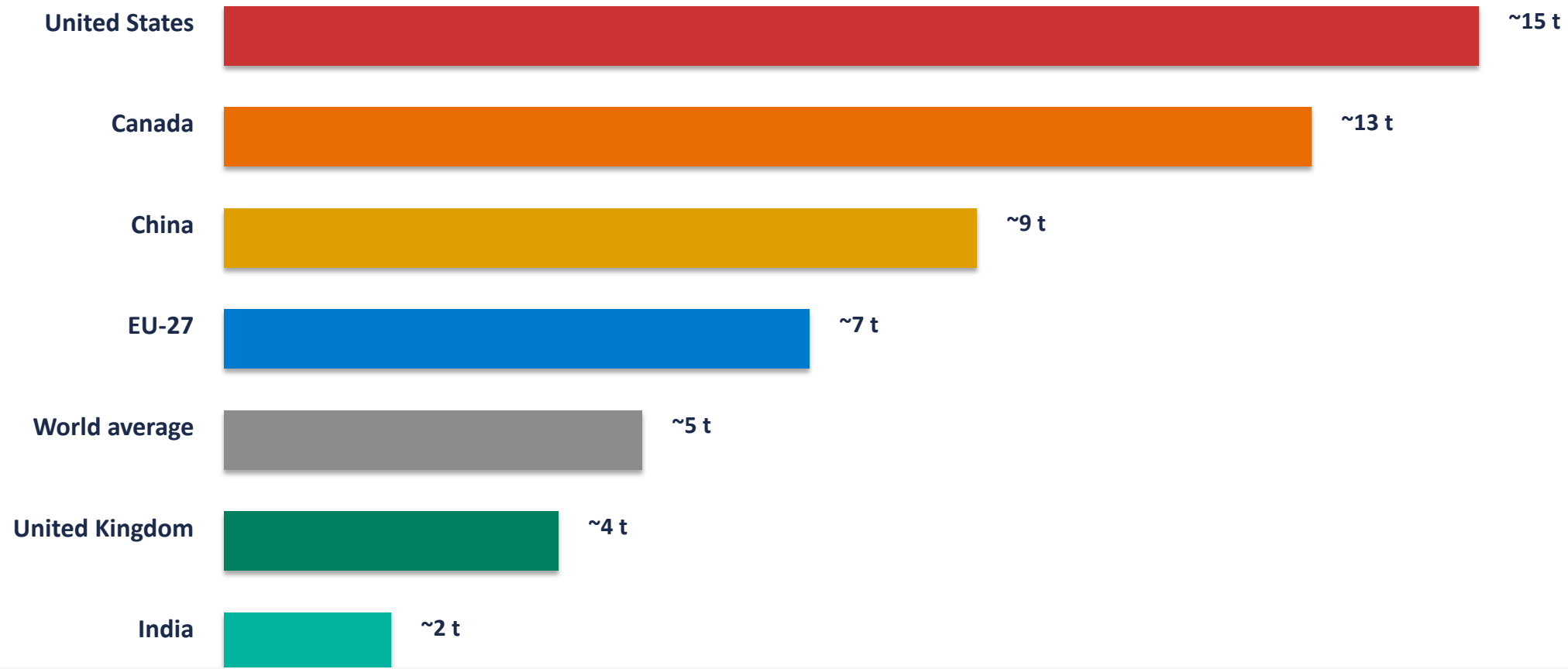


2024: Emission Shares



Asia is home to nearly 60% of the world's population. Africa and South America each contribute only 3–4%, comparable to international aviation and shipping combined.

Per Capita Gap: 15 t in the US vs. 0.1 t in Chad — a 150× Disparity



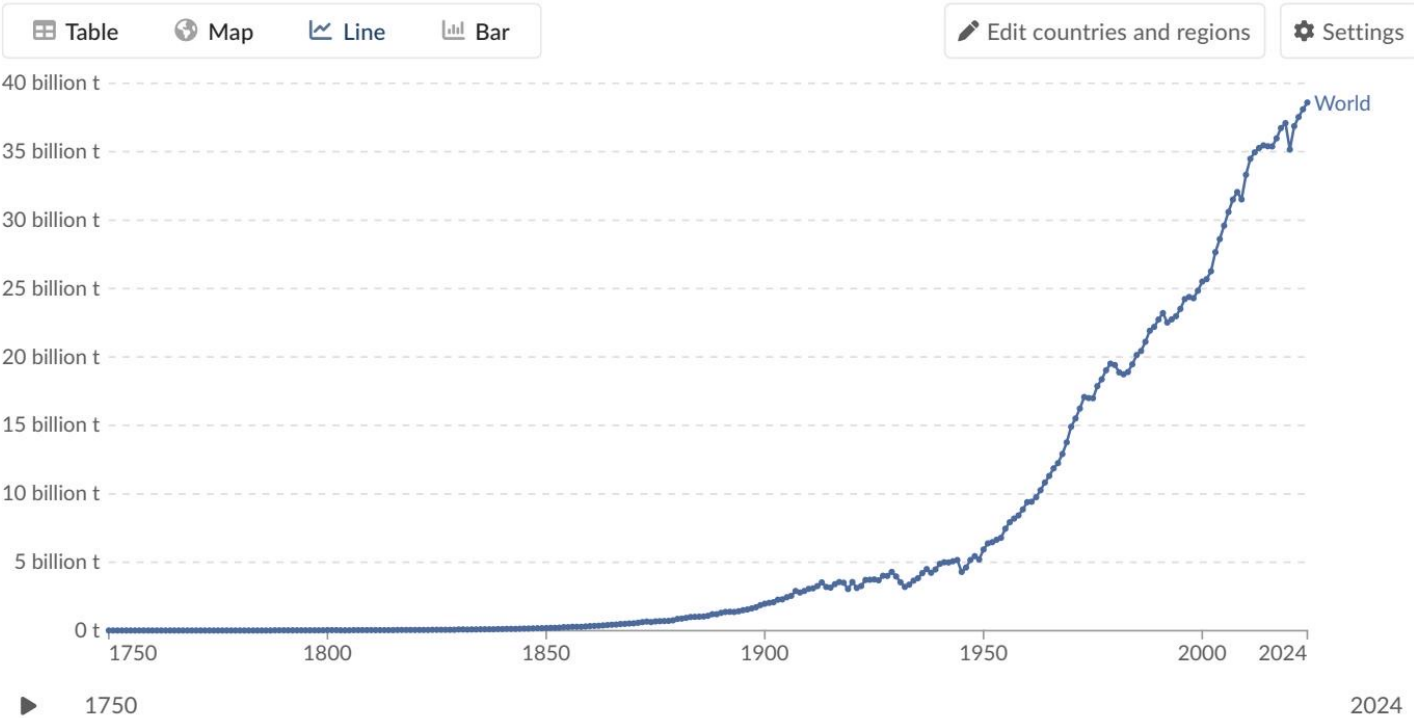
The average American emits in under two days what the average person in Chad emits in an entire year. | Reference: figure_1.jpg (per capita line chart)

2024 Snapshot: US and Europe Cut Emissions While Russia and SE Asia Increase

Annual CO₂ emissions

Carbon dioxide (CO₂) emissions from fossil fuels and industry. Land-use change emissions are not included.

Our World
in Data



Data source: Global Carbon Budget (2025) – [Learn more about this data](#)

OurWorldinData.org/co2-and-greenhouse-gas-emissions | CC BY



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figure_4.jpg: Annual % change in CO₂ emissions, 2024

Declining (–1% to –5%): US and much of Europe show blue, reflecting ongoing transitions away from coal and efficiency gains.

Increasing (+2% to +10%+): Russia, parts of Southeast Asia, and several African nations show orange/red.

Mixed in South America: Some countries experienced very large percentage increases (>10%), while others saw declines.

Overall: Global emissions growth is slowing but has not turned negative — aggregate emissions have not yet peaked.



Blue: Declining



Orange/Red: Increasing

Energy Mix Explains Why France Emits Far Less Per Capita Than Germany

Countries with similar living standards can have very different per capita emissions depending on their energy sources. France and the UK have much lower per capita emissions than Germany or the Netherlands, largely because of their energy mix — specifically, higher nuclear and renewable shares.

	France	United Kingdom	Germany	Netherlands
Primary Energy	Nuclear + renewables dominant	Gas + renewables + nuclear	Coal + gas + growing renewables	Gas-heavy + some renewables
Per Capita CO ₂	~4 t/year	~4 t/year	~7+ t/year	~8+ t/year
Key Factor	High nuclear share (~70% of electricity)	Coal phase-out + offshore wind	Continued coal use; late nuclear exit	Gas dependency; industrial base

Takeaway: Policy and energy mix choices — not just prosperity — determine per capita outcomes. A low-carbon grid is the single biggest lever for reducing national per capita emissions.

Key Takeaways: Emissions Are Still Rising, But the Map of Responsibility Is Shifting

1. Global CO₂ emissions now exceed 35 billion tonnes per year and have not yet peaked, despite slowing growth rates.
2. China alone accounts for over 25% of global emissions (~12.5 Bt), more than double the United States (~5 Bt); India (~3 Bt) is third-largest and rising fast.
3. **The geographic center of emissions has shifted dramatically:** from >90% in Europe and the US in 1900 to Asia producing ~50% today.
4. Per capita inequalities are extreme — a 150× gap separates the highest emitters (US, Australia at ~15 t/person) from the lowest (Chad, Niger at ~0.1 t/person).
5. **Energy mix and policy choices matter:** countries with similar prosperity levels (e.g., France vs. Germany) can have very different per capita emissions depending on their reliance on nuclear, renewables, or fossil fuels.